Students Performance Dashboard

- **Presented By**

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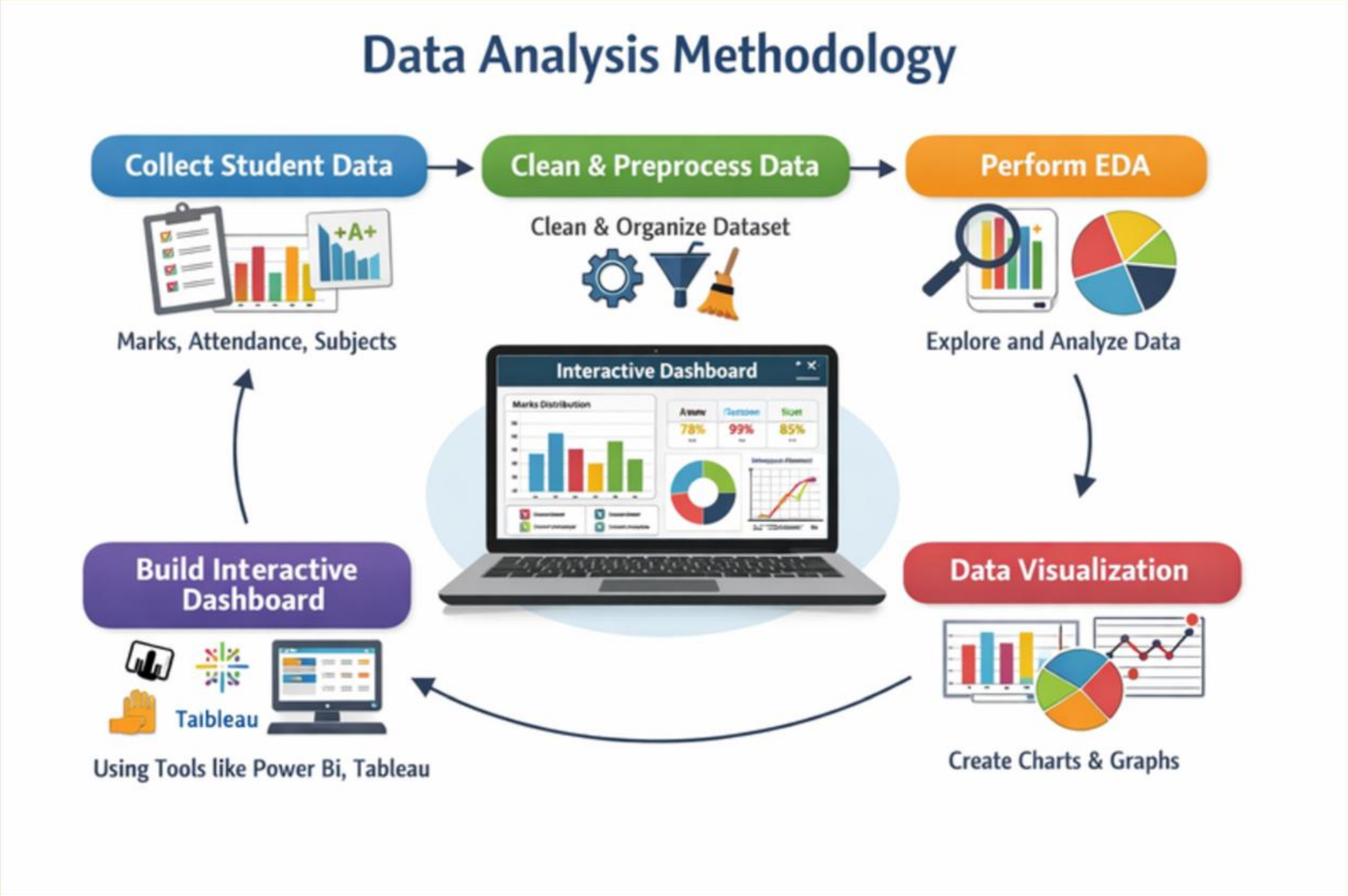
• INTRODUCTION

The Student Performance Dashboard is a data visualization system designed to analyze and monitor student performance effectively. It helps in tracking students' academic progress in a structured and efficient way. By using charts and graphs, the dashboard presents complex data in a simple and easy-to-understand format. This enables teachers and institutions to quickly identify trends, strengths, and areas that need improvement. Overall, the system supports data-driven decision making in education, making the evaluation process more accurate and efficient.

• OBJECTIVES

- To analyze student academic performance
- To identify strengths and weak areas of students
- To visualize data using interactive dashboards
- To help teachers make better decisions
- To improve overall student performance

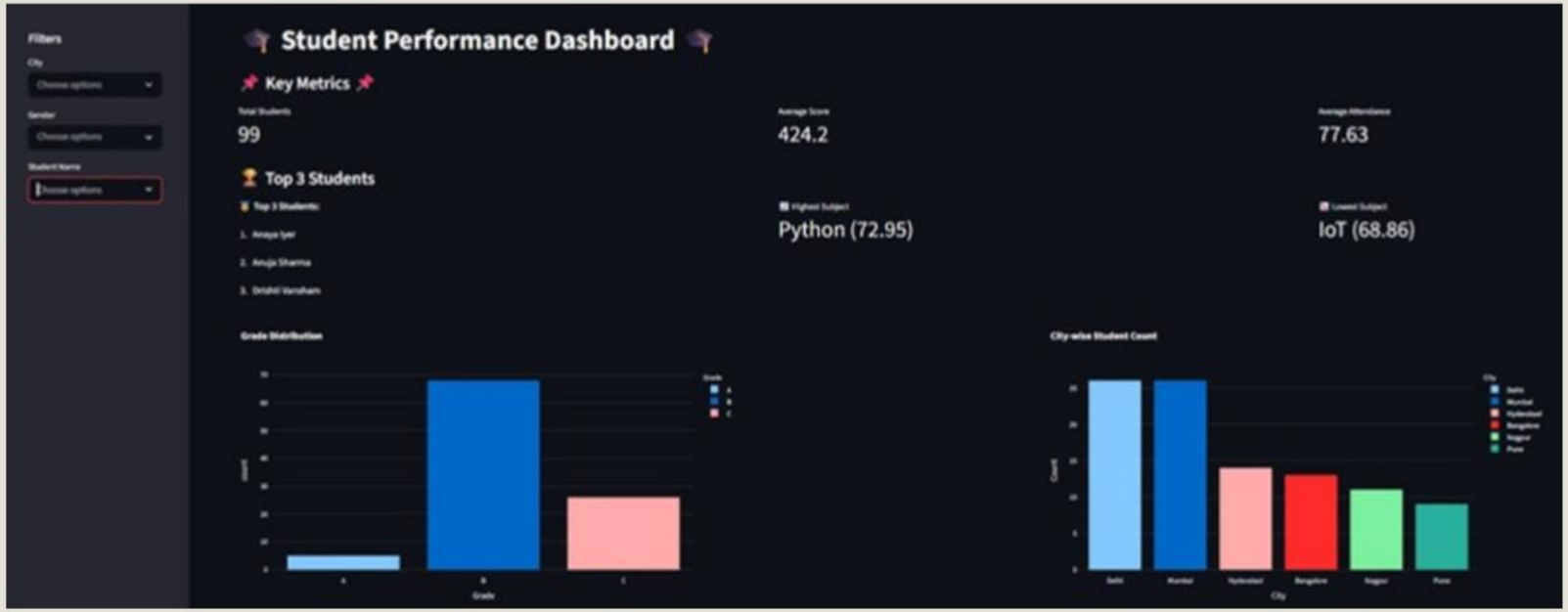
• Methodology

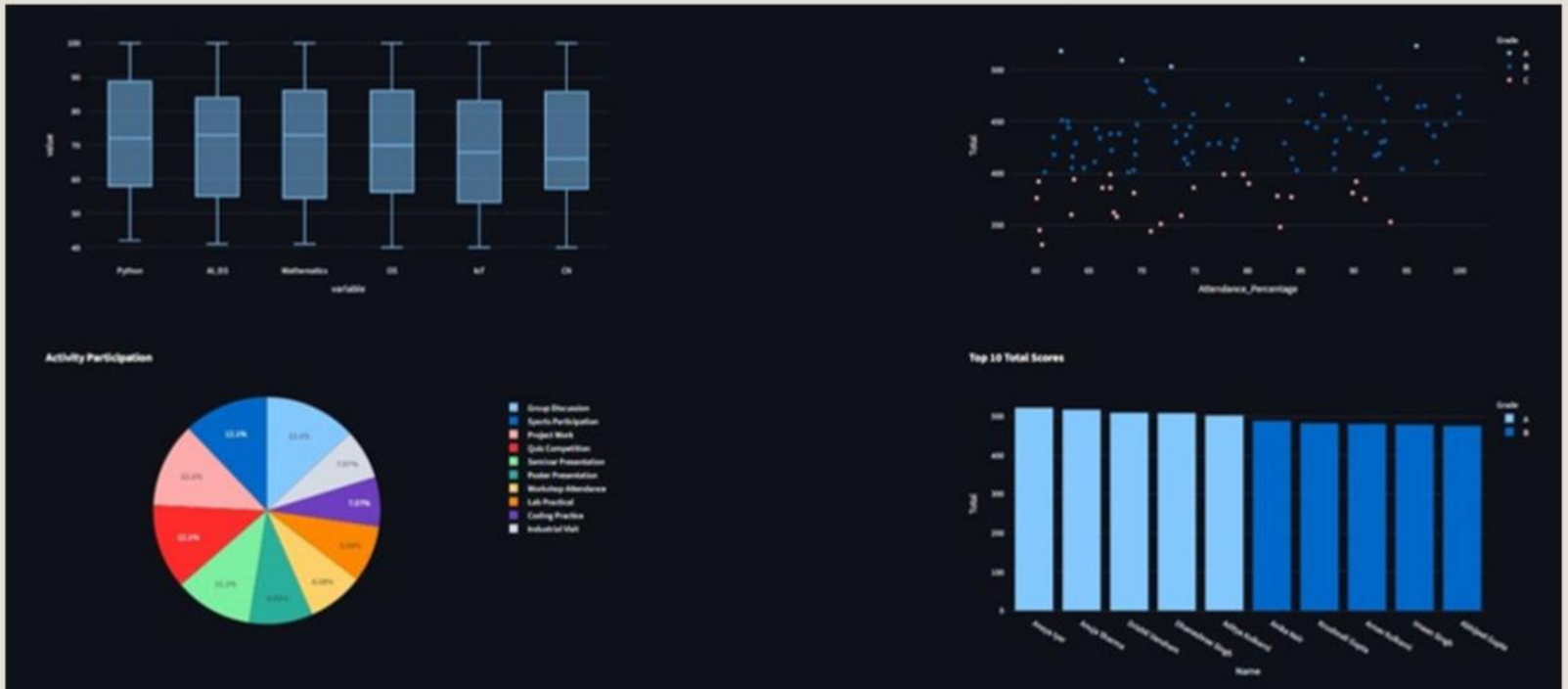


• TECHNICAL STACK

- Python – Data analysis and processing
- Pandas – Data cleaning and manipulation
- NumPy – Numerical computations
- Streamlit- Creating dashboard of UI
- Plotly – Interactive charts and graphs
- CSV Handling – Data storage and management
- Data Analysis Techniques (EDA) – Understanding patterns and trends

- **Results / Screenshots**





- Results / Screenshots

```
📥 Loading CSV file...
Original shape: (103, 17)
Removing empty rows...
After removing empty rows: (101, 17)
Removing rows with missing names...
After removing missing names: (99, 17)
Fixing column names...
Converting numeric columns: ['Python', 'AI_DS', 'Mathematics', 'OS', 'IoT', 'CN', 'Total', 'Attendance_Percentage']
Converting done.
Filling missing numeric values with mean...
Filling done.
Removing duplicate rows...
After removing duplicates: (99, 17)
✅ Preprocessing completed!
```

• EXPECTED OUTCOME

- Better understanding of student performance
- Easy identification of weak students
- Improved academic planning
- Time-saving for teachers
- Interactive and user-friendly dashboard

• CONCLUSION

- Dashboard simplifies performance analysis
- Helps in data-driven educational decisions
- Enhances student monitoring process
- Can be extended with AI predictions
- Useful for teachers, students, and institutions

THANK YOU

For Your Valuable Time